

File descriptors 1 :

```
#include <unistd.h>

int main()
{
    write(STDOUT_FILENO, "Hello World\n", 12);
    return 0;
}
```

```
#include <unistd.h>

int main()
{
    char buffer[20];

    read(STDIN_FILENO, buffer, sizeof(buffer));
    write(STDOUT_FILENO, buffer, sizeof(buffer));

    return 0;
}
```

```
bhanu@DESKTOP-DE32HQ8:~/file_permissions$ mkdir -p ~/file_descriptors
cd ~/file_descriptors
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ pwd
/home/bhanu/file_descriptors
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ pwd
/home/bhanu/file_descriptors
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano stdout.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc stdout.c -o stdout
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./stdout
Hello World
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano stdout.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano stdin.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc stdin.c -o stdin
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./stdin
♦♦♦Z7♦♦bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./stdin
Hello
Hello
♦: !♦~pXgbhanu@DESKTOP-DE32HQ8:~/file_descriptors$ |
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    char buffer[100];
    int n;

    printf("Enter something: ");

    n = read(STDIN_FILENO, buffer, sizeof(buffer) - 1);

    if (n > 0)
    {
        buffer[n] = '\0';
        write(STDOUT_FILENO, buffer, n);
    }

    return 0;
}
```

```
#include <unistd.h>

int main()
{
    write(STDERR_FILENO, "This is an error\n", 17);
    return 0;
}
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;

    fd = open("doesnotexist.txt", O_RDONLY);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    close(fd);

    return 0;
}
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;

    fd = open("student.txt", O_CREAT | O_WRONLY, 0644);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    printf("File created successfully!\n");
    printf("File descriptor = %d\n", fd);

    close(fd);

    return 0;
}
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;

    fd = open("student.txt", O_WRONLY | O_TRUNC);

    if (fd == -1)
    {
        perror("open");
        return 1;
    }

    write(fd, "New Data\n", 9);

    close(fd);

    return 0;
}
```

```
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ pwd
/home/bhanu/file_descriptors
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano stdout.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc stdout.c -o stdout
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./stdout
Hello World
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano stdout.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano stdin.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc stdin.c -o stdin
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./stdin

♦♦♦Z7♦♦bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./stdin
Hello
Hello
♦: !♦~pXgbhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano program3.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc program3.c -o program3
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./program3
Hello
Hello
Enter something: bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano stderr.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc stderr.c -o stderr
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./stderr
This is an error
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano error.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc error.c -o error
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./error
open: No such file or directory
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano create.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc create.c -o create
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./create
File created successfully!
File descriptor = 3
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ls -l student.txt
-rw-r--r-- 1 bhanu bhanu 0 Aug 29 04:02 student.txt
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano truncate.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc truncate.c -o truncate
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./truncate
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ cat student.txt
New Data
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ |
```

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd1, fd2;

    fd1 = open("student.txt", O_CREAT | O_WRONLY, 0644);

    if (fd1 == -1)
    {
        perror("open");
        return 1;
    }

    printf("fd1 = %d\n", fd1);

    fd2 = dup(fd1);

    printf("fd2 = %d\n", fd2);

    close(fd1);
    close(fd2);
}

```

New Data

```

bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ nano dup.c
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ gcc dup.c -o dup
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ ./dup
fd1 = 3
fd2 = 4
bhanu@DESKTOP-DE32HQ8:~/file_descriptors$ |

```